

Land Survey Analysis

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Computational Geometry

Universe

- Building Footprint: $\{width, height, center_x, center_y\}$
- Max Inscribed Rectangle: $\{center_x, center_y, width, height, area_m2\}$

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

A surveyor has recorded GPS coordinates for 8 boundary markers defining an irregular lot. Given these coordinates (in meters, local projected coordinate system), compute:

- Area** of the lot polygon
- Perimeter** of the lot boundary
- Convex hull** of the survey points
- Centroid** (geometric center of mass)
- Building fit check** -- determine whether a proposed rectangular building footprint fits entirely within the lot

The lot is an irregular 8-sided polygon. The building footprint is an axis-aligned rectangle placed at a proposed location. The solver must verify containment using independent geometric methods.

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) Computational geometry *algorithms operating on points, lines, and polygons*

- (2) Polygon operations
- (3) Spatial containment
- (4) Voronoi partitioning
- (5) Shoelace formula
- (6) Convex hull

area, perimeter, convex hull, centroid for a polygon
testing whether one geometric object lies entirely within another
dual of Delaunay triangulation, useful for nearest neighbor queries
exact $O(n)$ area computation for simple polygons
smallest convex set containing all points; many applications

Approach

Suggested approach Shoelace + ConvexHull(Qhull) + Shapely containment

Complexity class: $O(n)$

Available tools: Shoelace + ConvexHull

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	Shoelace + ConvexHull(Qhull) + Shapely containment $O(n)$
Time:	1.8648s

Solution Details

Method: 1. Area: Shoelace formula (Gauss's area formula) for simple polygons -- $O(n)$
2. Perimeter: Sum of Euclidean distances between consecutive vertices -- $O(n)$
3. Convex hull: Scipy ConvexHull (Qhull library, $O(n \log n)$)
4. Centroid: Weighted average using the shoelace sub-expressions -- $O(n)$
5. Containment: Shapely Polygon.contains() for the full building rectangle -- $O(n)$ per query
6. Max inscribed rectangle: Grid search over candidate positions and sizes within the polygon

Verification

✓ Feasibility All constraints satisfied

3. Interpretation

The Question

A surveyor has recorded GPS coordinates for 8 boundary markers defining an irregular lot.

The Answer

A feasible solution was found.

What This Means

- Lot area computed via shapely and independently verified with the shoelace formula
- Perimeter of the lot boundary
- Convex hull vertices and hull area (compared to lot area for concavity measure)
- Centroid coordinates
- Building containment verdict (fits / does not fit)
- Maximum axis-aligned inscribed rectangle estimate
- Independent verification of all computed quantities

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Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution